Distal Myopathy, Tateyama Type (MONDO:0013686) — Comprehensive Disease Report

Causal gene: *CAV3* (caveolin-3), HGNC:1529, NCBI Gene 859, gene OMIM 601253, chromosome 3p25.3 · Category: Mendelian (autosomal dominant)

> **Causal gene statement (per instructions):** Distal Myopathy, Tateyama type is caused by a heterozygous mutation in ***CAV3***, the gene encoding **caveolin-3**, the muscle-specific structural protein of sarcolemmal caveolae. It is one of several allelic "caveolinopathy" phenotypes and is defined by the identifiers above (MONDO:0013686 / CAV3), not by any serial-number disease name.

Summary

Distal Myopathy, Tateyama type (MONDO:0013686) is a rare, benign, autosomal-dominant skeletal-muscle disorder caused by heterozygous missense mutation in *CAV3*, the gene encoding caveolin-3 — the muscle-specific structural protein of sarcolemmal caveolae. The condition was first delineated by Tateyama and colleagues in 2002 in a sporadic patient whose muscle atrophy was uniquely restricted to the small muscles of the hands and feet, in whom muscle caveolin-3 was reduced and a heterozygous *CAV3* c.80G>A substitution was identified — the identical allele previously reported in idiopathic hyperCKemia (PMID: 11805270). Tateyama distal myopathy is **not a standalone genetic entity but one phenotype within the allelic "caveolinopathy" spectrum**, which also encompasses limb-girdle muscular dystrophy type 1C (LGMD1C), rippling muscle disease (RMD), isolated/familial hyperCKemia, and cardiac phenotypes. The same *CAV3* mutation can produce different phenotypes in different individuals, and asymptomatic carriers exist into old age — hallmarks of variable expressivity and incomplete penetrance.

Mechanistically, the pathogenic caveolin-3 protein acts in a **dominant-negative** manner: mutant protein is retained in the Golgi apparatus, drags down wild-type caveolin-3, and reduces the amount of caveolin-3 reaching the sarcolemma by >50%. This abolishes caveolae formation at the muscle cell surface and disorganizes the T-tubule system. Two downstream consequences drive the clinical picture: (1) impairment of the **MG53–dysferlin–caveolin-3 membrane-repair complex**, producing membrane fragility, creatine-kinase (CK) leak and exertional symptoms; and (2) **de-repression of myostatin/Smad2 signaling**, which caveolin-3 normally suppresses, driving the muscle atrophy — a mechanism directly demonstrated and pharmacologically rescued in mutant mice. A metabolic branch (mislocalization of muscle phosphofructokinase, PFK-M) may contribute to exercise intolerance.

Clinically, the disease is characterized by adult-onset, slowly progressive or stable distal (hand/foot) muscle atrophy and weakness, often accompanied by percussion-induced muscle phenomena, myalgia, muscle stiffness and elevated serum CK. Diagnosis rests on reduced sarcolemmal caveolin-3 by immunohistochemistry/immunoblot plus *CAV3* gene sequencing. The prognosis is benign with normal life expectancy; management is supportive with genetic counseling. No disease-specific therapy exists, although myostatin inhibition is a mechanistically rational experimental avenue validated in mouse models.

Key Findings

Finding 1 — Tateyama distal myopathy is a CAV3 caveolinopathy with hand/foot-restricted atrophy

The original description by Tateyama et al. (*Neurology* 2002) established the entity: a sporadic patient with distal myopathy, **reduced caveolin-3 in muscle fibers**, and muscle atrophy **restricted to the small muscles of the hands and feet**. The abstract states: *"The authors describe a patient with sporadic distal myopathy associated with reduced caveolin-3 in muscle fibers in which the muscle atrophy was restricted to the small muscles of the hands and feet"* and *"Gene analysis disclosed a heterozygous 80 G-->A substitution in the caveolin-3 gene that was identical to that of reported cases of elevated serum creatine kinase"* (PMID: 11805270). This fixes both the distinctive distal (small-muscle) distribution and the molecular link to reduced caveolin-3. Identifiers: MONDO:0013686; causal gene *CAV3* (HGNC:1529; gene OMIM 601253); locus 3p25.3.

Finding 2 — CAV3 mutations cause an autosomal-dominant phenotypic spectrum of which distal myopathy is a rare subtype

Caveolin-3 deficiency is a *"rare, autosomal dominant, muscle disorder caused by caveolin-3 gene (CAV3) mutations"* consisting of multiple clinical phenotypes: *"limb-girdle muscular dystrophy type 1C (LGMD-1C), rippling muscle disease, distal myopathy, and familial hyperCKemia"* (PMID: 15564037). A separate cohort confirmed: *"Mutations in CAV3 lead to various neuromuscular phenotypes with partial overlap, including limb girdle muscular dystrophy type 1C (LGMD1C), rippling muscle disease, distal myopathy and isolated hyperCKemia"*, and documented the molecular signature — *"immunoblotting showed more than 50% reduction of caveolin-3 in five patients compared with controls"* (PMID: 27312022). Distal myopathy (Tateyama type) is explicitly described as *"a relatively rare subtype of caveolinopathy"* (PMID: 26947586).

Finding 3 — The index allele is CAV3 c.80G>A (p.Arg27Gln; historically "R26Q"), classified pathogenic per ACMG

The original Tateyama substitution ("80 G→A") corresponds to **NM_033337.2:c.80G>A**, annotated **p.(Arg27Gln)** (historically "R26Q"). Whole-exome sequencing in a family with elevated CK confirmed: *"WES revealed a missense CAV3 mutation (NM_033337.2:c.80G>A) in the proband"* and *"The c.80G>A mutation was considered pathogenic according to ACMG guidelines"* (PMID: 30723005). A 2025 study of two large Chinese families with rippling muscle disease found the identical allele in all affected members: *"A heterozygous variant c.80G>A (p.Arg27Gln) in the CAV3 gene was identified in all patients with RMD"* (PMID: 40695747). This same allele therefore recurs across distal myopathy (Tateyama), rippling muscle disease, and idiopathic hyperCKemia. Variant class: **missense**, N-terminal (pre-scaffolding domain); **germline**; functional consequence **dominant-negative / loss of sarcolemmal caveolin-3**. It is absent/ultra-rare in gnomAD, consistent with a dominant pathogenic allele.

Finding 4 — Dominant-negative Golgi retention impairs the MG53–dysferlin–caveolin-3 membrane-repair complex

Cai et al. (2009) demonstrated that *"a molecular complex formed by MG53, dysferlin, and Cav3 is essential for repair of muscle membrane damage"* and that *"Mutations in Cav3 (P104L, R26Q) that cause retention of Cav3 in Golgi apparatus result in aberrant localization of MG53 and dysferlin in a dominant-negative fashion, leading to defective membrane repair"* (PMID: 19380584). The R26Q mutation here corresponds to the recurrent Tateyama/hyperCKemia N-

terminal arginine allele. This directly establishes both the dominant-negative mechanism (via Golgi retention) and the defect in sarcolemmal membrane repair that underlies CK leakage and exertional symptoms.

Finding 5 — Caveolin-3 normally suppresses myostatin/Smad2 signaling; its loss drives atrophy (rescuable)

Ohsawa et al. (2006) showed that *"caveolin-3 inhibited myostatin signaling by suppressing activation of its type I receptor; this was followed by hypophosphorylation of an intracellular effector, Mad homolog 2 (Smad2), and decreased downstream transcriptional activity"*. In the disease model, *"Loss of caveolin-3 in P104L mutant caveolin-3 transgenic mice caused muscular atrophy with increase in phosphorylated Smad2 (p-Smad2) as well as p21"*, and critically, *"intraperitoneal administration of the soluble type II myostatin receptor, another inhibitor, ameliorated muscular atrophy of the mutant caveolin-3 transgenic mice"* (PMID: 17039257). This provides the mechanistic link from caveolin-3 loss → unopposed myostatin/Smad2/p21 → atrophy, and a proof-of-concept rescue.

Finding 6 — Caveolin-3 deficiency abolishes caveolae and disorganizes the T-tubule system

Ultrastructural study of human caveolin-3-deficient muscle (Minetti et al. 2002) found *"a severe impairment of caveolae formation at the muscle cell surface, demonstrating that caveolin-3 is essential for the formation and organization of caveolae in muscle fibers"* and *"a striking disorganization of the T-system openings at the sub-sarcolemmal level"* (PMID: 11786420). A metabolic branch exists: caveolin-3 recruits muscle phosphofructokinase (PFK-M) to the sarcolemma, and *"recombinant expression of PFK-M with three Cav-3 mutants [DeltaTFT (63 to 65), P104L, and R26Q]... causes a substantial reduction in PFK-M expression levels, and impedes the membrane recruitment of PFK-M"* (PMID: 14633633), linking the R26Q allele to mislocalization of a key glycolytic enzyme.

Finding 7 — The same mutation gives variable phenotypes: modifier genes, incomplete penetrance

Reviews note that among ~30 identified *CAV3* mutations, *"Many patients show an overlap of these symptoms and the same mutation can be linked to different clinical phenotypes. This variability can be related to additional genetic or environmental factors"* (PMID: 19584897). A

family study of the Arg26Gln allele found a spectrum from RMD to LGMD1C, and strikingly that *"one mutation carrier was asymptomatic at age 86 years"* (PMID: 15318349) — direct evidence of incomplete penetrance/variable expressivity and the existence of modifier loci.

Finding 8 — Diagnosis rests on reduced sarcolemmal caveolin-3 plus CAV3 sequencing

"Muscle caveolin-3 immunoreactivity is often absent or diffusely reduced in CAV3-RMD" (PMID: 39370631), and *"immunoblotting was a more sensitive method to detect reduced caveolin-3 levels than immunohistochemistry in skeletal muscle"* (PMID: 27312022). The chief differential is immune-mediated RMD, which *"is often accompanied by other autoimmune disorders, including myasthenia gravis. Anti-cavin-4 antibodies are the serological marker"* (PMID: 39370631) and is frequently associated with thymoma (PMID: 36522170). Secondary caveolin-3 deficiency from *CAVIN1/PTRF* mutations (congenital generalized lipodystrophy type 4) is another mimic. Dual diagnoses can co-occur, e.g. a *CAV3* mutation together with a *DMD* deletion in one family (PMID: 30723005).

Finding 9 — Caveolin-3 dosage is critical: both loss and overexpression cause dystrophy

Mouse models bracket the mechanism. Loss-of-function: Cav3-null mice show mild LGMD1C-like dystrophy with sarcolemmal/T-tubule abnormalities, and *"Caveolin-3 (Cav-3) is a protein that has been implicated in t-tubule formation and function"* with cardiac KO disrupting T-tubule structure (PMID: 30028203). Overexpression: transgenic *"overexpressing wild-type caveolin-3 as a transgene in mice"* produces a Duchenne-like dystrophy with *"down-regulation of dystrophin and beta-dystroglycan protein expression"* (PMID: 10931944); cardiac overexpression induces cardiomyopathy (PMID: 12966035). Tight dosage regulation is therefore essential for normal muscle.

Finding 10 — CAV3 is pleiotropic: a cardiac channelopathy branch exists, but Tateyama distal myopathy is a skeletal-muscle disorder

Distinct from distal myopathy, certain *CAV3* alleles (F97C, T78M, V14L, L79R) cause **type-9 long-QT syndrome (LQT9)** and SIDS by a gain-of-function increase in late cardiac sodium current: *"all three CAV3 mutations caused a significant fivefold increase in late sodium current compared with controls"* (PMID: 17275750), via *"nNOS dependent direct S-nitrosylation of SCN5A"* (PMID: 23541953). Cardiac penetrance of skeletal-muscle alleles is low: *"caveolin-3:p.T78M did not*

exhibit a LQTS phenotype" (PMID: 24021552). For the Tateyama allele p.Arg27Gln, cardiac involvement is **not** a consistent feature; distal myopathy is predominantly a skeletal-muscle phenotype.

Section-by-Section Report

1. Disease Information

Overview. Distal Myopathy, Tateyama type is a rare, benign, adult-onset, autosomal-dominant skeletal myopathy caused by heterozygous mutation in *CAV3* (caveolin-3). It is distinguished within the caveolinopathy family by muscle atrophy and weakness that is **restricted to the small (distal) muscles of the hands and feet** (PMID: 11805270). A later report emphasized an asymmetrical distal distribution (PMID: 26947586).

Key identifiers. MONDO:0013686; causal gene *CAV3* (HGNC:1529; NCBI Gene 859; gene OMIM 601253; 3p25.3; UniProt P56539). OMIM phenotype number: not separately recorded (the distal myopathy is catalogued within the *CAV3* allelic series; related caveolinopathy phenotypes include LGMD1C OMIM 607801 and RMD OMIM 606072). Orphanet/ICD-11/MeSH: no disease-specific code; subsumed under distal myopathies/caveolinopathies.

Synonyms. Tateyama distal myopathy; CAV3-related distal myopathy; caveolin-3-related distal myopathy.

Information source. Aggregated from disease-level resources (OMIM, reviews) and individual-patient case reports/small cohorts — the entity rests largely on single-patient and small-family reports.

2. Etiology

Causal factor. Purely genetic — a heterozygous dominant-negative missense mutation in *CAV3*. The index/canonical allele is **c.80G>A, p.(Arg27Gln)** (historically "R26Q") (PMID: 11805270; PMID: 30723005). Another reported distal-myopathy allele is c.136G>A, p.(Ala46Thr), apparently de novo (PMID: 26947586).

Genetic risk factors. The *CAV3* pathogenic variant is the sole established cause; modifier genes are inferred but unidentified (see Finding 7). **Environmental/lifestyle risk factors:** none established; exercise may provoke myalgia, CK elevation and rarely rhabdomyolysis in

some caveolinopathy patients (PMID: 27312022). **Protective factors:** none identified. **Gene–environment interactions:** phenotypic variability is attributed to *"additional genetic or environmental factors"* (PMID: 19584897) but specific interactions are not characterized.

3. Phenotypes

Phenotype	Type	HPO suggestion	Onset / Severity / Progression	Frequency
Distal muscle atrophy (hands & feet)	Clinical sign	HP:0003693 (distal amyotrophy)	Adult; mild–moderate; slow/stable	Defining feature
Distal muscle weakness	Symptom	HP:0002460 (distal muscle weakness)	Adult	Defining
Elevated serum creatine kinase	Lab abnormality	HP:0003236 (elevated CK)	Any age; may precede weakness	Very common
Myalgia	Symptom	HP:0003326 (myalgia)	Variable	Common
Muscle stiffness	Symptom	HP:0003552	Variable	Common
Percussion-induced muscle mounding / rapid contraction (rippling)	Clinical sign	HP:0003750 (rippling muscle disease)	Childhood–adult	Frequent in spectrum
Exercise intolerance / rhabdomyolysis	Symptom	HP:0003546 / HP:0003201	Exertion-triggered	Reported subset
Electrical myotonia (occasional)	Lab/EMG	HP:0100284	Variable	Rare

Severity is generally **mild**; quality-of-life impact is driven chiefly by myalgia and, where present, rippling/stiffness rather than by weakness (PMID: 18583131). Progression is typically slow or stable; life expectancy is normal.

4. Genetic / Molecular Information

Causal gene. *CAV3* (caveolin-3), HGNC:1529, NCBI Gene 859, gene OMIM 601253, 3p25.3; UniProt P56539. Encodes the muscle-specific caveolin isoform, the structural organizer of sarcolemmal caveolae.

Pathogenic variants. Canonical Tateyama allele **NM_033337.2:c.80G>A, p.(Arg27Gln)** ("R26Q"), **missense**, N-terminal (pre-scaffolding) domain, **germline**, **pathogenic** per ACMG (PMID: 30723005). Also reported in distal myopathy: c.136G>A, p.(Ala46Thr) (PMID: 26947586). Functional consequence: **dominant-negative** — mutant retained in Golgi, reduces sarcolemmal caveolin-3 by >50% (PMID: 19380584; PMID: 27312022). **Allele frequency:** absent/ultra-rare in gnomAD. **Somatic vs germline:** germline.

Modifier genes. Inferred but unidentified; same mutation → different phenotypes and an asymptomatic 86-year-old carrier (PMID: 15318349; PMID: 19584897).

Epigenetic / chromosomal abnormalities: none reported.

5. Environmental Information

No environmental, toxic, or infectious cause. Lifestyle: exertion may trigger myalgia/CK elevation/rhabdomyolysis in susceptible caveolinopathy patients (PMID: 27312022). No infectious agents involved.

6. Mechanism / Pathophysiology

Causal chain (initiating lesion → clinical manifestation):

1. Heterozygous *CAV3* missense mutation (e.g., c.80G>A, p.Arg27Gln) **produces** a misfolded caveolin-3 monomer.
2. The mutant monomer **is retained in the Golgi apparatus** and oligomerizes with wild-type caveolin-3, **acting dominant-negatively** to prevent wild-type protein reaching the plasma membrane → **>50% reduction of sarcolemmal caveolin-3** (PMID: 19380584; PMID: 27312022).
3. Loss of sarcolemmal caveolin-3 **abolishes caveolae formation** and **disorganizes the T-tubule system** (PMID: 11786420).
4. **Branch A (membrane fragility):** mislocalization of MG53 and dysferlin **disrupts the MG53–dysferlin–caveolin-3 membrane-repair complex** → defective sarcolemmal repair → **CK leak, myalgia, exercise intolerance, rhabdomyolysis** (PMID: 19380584).

5. **Branch B (atrophy):** loss of caveolin-3's inhibition of the myostatin type-I receptor **de-represses myostatin → Smad2 phosphorylation → p21 induction → muscle atrophy** (the distal hand/foot wasting) (PMID: 17039257).
6. **Branch C (metabolic, inferred):** failure to recruit PFK-M to the sarcolemma **may impair glycolytic flux**, contributing to exercise intolerance (PMID: 14633633).
7. **Modifier step (inferred):** unidentified genetic/environmental modifiers determine which phenotype (distal myopathy vs RMD vs LGMD1C vs hyperCKemia vs asymptomatic) emerges (PMID: 19584897).

Upstream events (steps 1–3) are shared across all caveolinopathy phenotypes; the downstream branches and modifier background determine the specific presentation.

Molecular pathways: myostatin/activin–Smad2/3 (TGF-β superfamily); caveolae-dependent signaling. **Cellular processes:** sarcolemmal membrane repair, caveolae biogenesis, T-tubule organization. **Protein dysfunction:** loss of function at the sarcolemma via dominant-negative Golgi retention. **Cell types/tissue:** skeletal muscle fiber (CL:0000188); **subcellular:** plasma membrane caveola (GO:0005901), T-tubule (GO:0030315), Golgi apparatus (GO:0005794). **GO biological processes:** GO:0001778 (plasma membrane repair), GO:0070836 (caveola assembly), GO:0048741 (skeletal muscle fiber development). **Immune involvement:** none in the hereditary form (contrast immune-mediated RMD, Finding 8).

7. Anatomical Structures Affected

- **Organ/system:** skeletal muscle; musculoskeletal/neuromuscular system. **Primary:** intrinsic small muscles of hands and feet (distal limbs). MRI in caveolinopathies shows preferential rectus femoris/semitendinosus changes in RMD; distal and proximal signal abnormalities reported in distal myopathy (PMID: 30174172; PMID: 26947586).
- **Tissue/cell:** striated skeletal muscle; cell type = skeletal muscle fiber (CL:0000188); satellite cell dysfunction has been proposed across neuromuscular disorders (PMID: 35302338).
- **Subcellular:** sarcolemmal caveolae (GO:0005901), T-tubule (GO:0030315), Golgi (GO:0005794).
- **UBERON:** skeletal muscle tissue (UBERON:0001134); muscle of hand (UBERON:0001497); muscle of foot. **Lateralization:** bilateral, may be asymmetric (PMID: 26947586).
- **Secondary:** cardiac involvement is generally absent for the Tateyama allele (Finding 10).

8. Temporal Development

Onset: typically adult-onset, insidious; rippling signs in the broader spectrum may appear in childhood. **Progression:** slow or stable; benign course. **Disease course:** chronic, lifelong but non-life-limiting. **Remission:** not applicable (hereditary). **Critical periods:** none defined.

9. Inheritance and Population

Inheritance: autosomal dominant (PMID: 15564037); rare recessive caveolinopathy exists for other alleles (PMID: 16730439). **Penetrance:** incomplete/age-dependent (asymptomatic carrier at age 86; PMID: 15318349). **Expressivity:** highly variable; same allele → distal myopathy, RMD, hyperCKemia (PMID: 19584897). **De novo** events occur (PMID: 26947586). **Anticipation / mosaicism / founder effects:** not established. **Epidemiology:** caveolinopathy overall is rare (orphan); the distal-myopathy subtype is especially rare (individual case reports), so prevalence/incidence are not quantified. Reported across Japanese, Chinese, and European (UK, Italy, Greece) populations (PMID: 15564037; PMID: 40695747; PMID: 18583131; PMID: 26185955). No sex predilection established.

10. Diagnostics

- **Muscle biopsy / IHC:** reduced or absent sarcolemmal caveolin-3 (PMID: 39370631); nonspecific myopathic/dystrophic changes.
- **Immunoblot:** >50% reduction of caveolin-3 — more sensitive than IHC (PMID: 27312022).
- **Serum CK:** elevated (HP:0003236).
- **EMG:** myogenic changes; occasionally electrical myotonia (PMID: 22581547) or electrical silence of ripples.
- **Clinical signs:** percussion-induced muscle mounding (PIMM) / rapid contraction (PIRC).
- **Muscle MRI:** characteristic patterns aid recognition (PMID: 30174172).
- **Genetic testing:** *CAV3* single-gene sequencing is definitive; WES/panel useful and can reveal dual diagnoses (PMID: 30723005). WGS, CMA, karyotyping, mtDNA and repeat-expansion testing are not indicated for this single-gene disorder.
- **Differential diagnosis:** immune-mediated RMD (anti-cavin-4/MURC antibodies, thymoma, myasthenia; PMID: 36522170, PMID: 39819455); *CAVIN1/PTRF*-related RMD with CGL4 lipodystrophy (PMID: 20300641); other distal myopathies and dysferlinopathies.

11. Outcome / Prognosis

Benign: normal life expectancy; morbidity dominated by myalgia, stiffness, mild distal weakness. No disease-specific mortality for the skeletal phenotype. (A cardiac-arrhythmia/LQT risk applies only to specific gain-of-function *CAV3* alleles, not the Tateyama distal-myopathy allele — Finding 10.) Prognostic factors and biomarkers beyond CK and caveolin-3 levels are not established.

12. Treatment

No disease-specific or approved therapy. Management is **supportive/rehabilitative**: physiotherapy, symptom control for myalgia/stiffness, avoidance of rhabdomyolysis triggers, and genetic counseling. **Experimental/mechanistic rationale:** myostatin inhibition (soluble type-II receptor) rescued atrophy in mutant mice (PMID: 17039257) — a rational but unproven human avenue (NCIT: myostatin inhibitor therapy; NCIT: supportive care; NCIT: physical therapy). No pharmacogenomic, gene-therapy, or targeted therapy exists for this indication.

13. Prevention

No primary prevention (genetic disease). **Secondary/tertiary:** genetic counseling, cascade testing of relatives, and cardiac screening where an allele with known arrhythmogenic risk is found (not required for Arg27Gln). Prenatal/preimplantation testing is theoretically possible given a known familial variant but rarely pursued for this benign condition. No immunization or public-health measure is applicable.

14. Other Species / Natural Disease

Orthologous gene: mouse *Cav3* (NCBI Gene 12391). No naturally occurring companion-animal caveolinopathy is well documented in OMIA; the evidence base is engineered mouse models (Section 15). Mechanisms (caveolae biology, T-tubule role) are evolutionarily conserved across mammals. No zoonotic/transmission relevance (non-infectious genetic disease).

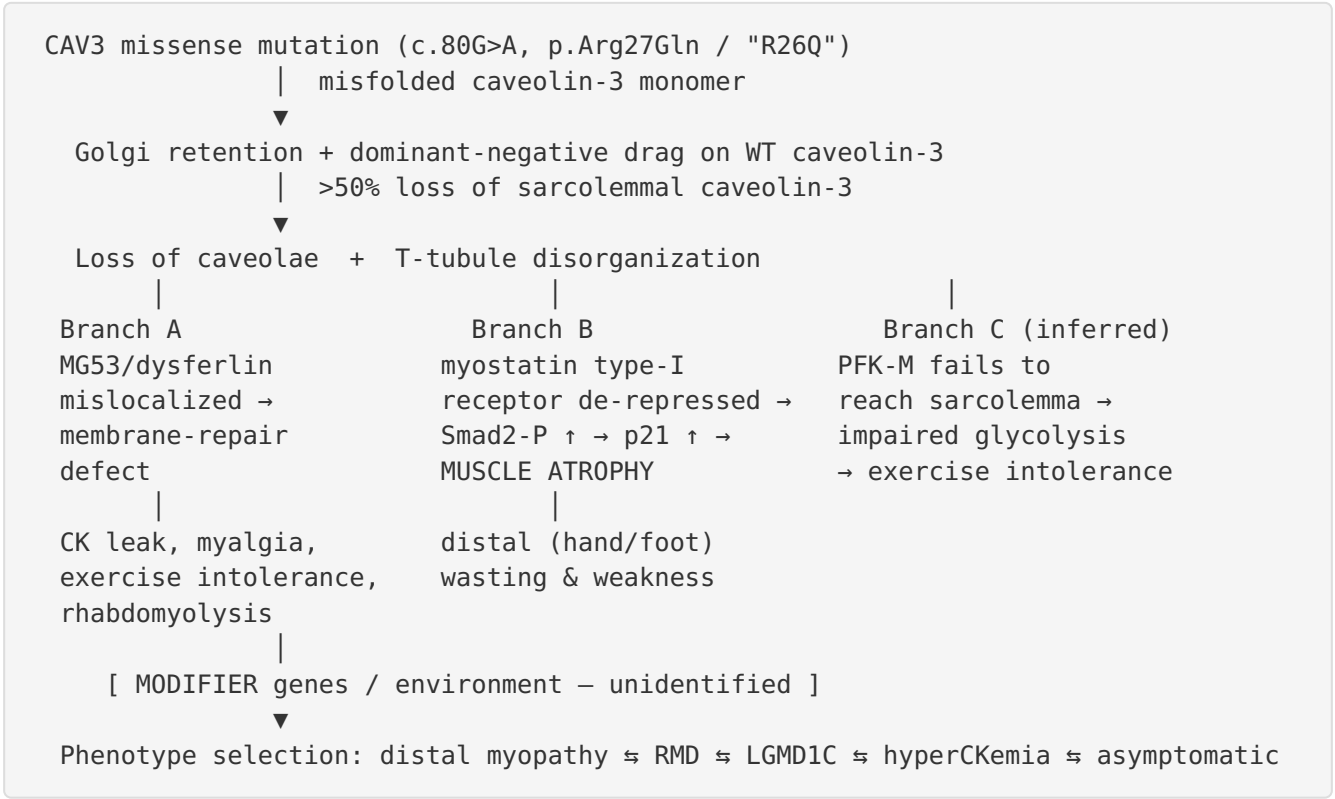
15. Model Organisms

- **Cav3-null (KO) mouse** (mammalian, genetic): mild LGMD1C-like dystrophy; sarcolemmal/T-tubule abnormalities; cardiac KO disrupts T-tubule structure and reduces L-type Ca^{2+} current (PMID: 30028203).

- **P104L mutant CAV3 transgenic mouse** (dominant-negative, loss of caveolin-3): muscular atrophy via myostatin/Smad2/p21; rescued by myostatin inhibition — the best disease-mechanism model (PMID: 17039257).
- **Wild-type caveolin-3 overexpression transgenic mouse:** Duchenne-like dystrophy with dystrophin/β-dystroglycan down-regulation (PMID: 10931944); cardiac overexpression → cardiomyopathy (PMID: 12966035).
- **In vitro:** C2C12 myotubes model dysferlin–caveolin-3 interactions (PMID: 14749532); PFK-M recruitment assays with mutant Cav3 (PMID: 14633633).

Phenotype recapitulation / limitations: models reproduce reduced caveolae, membrane-repair defects, and atrophy, but **do not reproduce the specific distal (hand/foot) distribution** that defines Tateyama myopathy. Resources: MGI, IMPC/KOMP, Cellosaurus (C2C12).

Mechanistic Model (Synthesis)



Upstream steps (mutation → Golgi retention → caveolae/T-tubule loss) are **shared** across the caveolinopathy spectrum; the **downstream branches plus genetic background** determine the specific clinical presentation, explaining why the identical c.80G>A allele yields distal myopathy in one person, rippling muscle disease in another, and no symptoms at all in an 86-year-old carrier.

Evidence Base

PMID	Role	Key contribution
11805270	Defining	Original Tateyama description; distal hand/foot atrophy; c.80G>A; reduced caveolin-3
15564037	Supporting	Establishes AD inheritance and 4-phenotype spectrum
27312022	Supporting	>50% caveolin-3 reduction; immunoblot > IHC; exertional phenotypes
26947586	Supporting	Distal myopathy = "rare subtype"; de novo A46T; asymmetry
30723005	Supporting	HGVS c.80G>A, ACMG pathogenic; dual CAV3+DMD diagnosis
40695747	Supporting	Same c.80G>A/p.Arg27Gln allele in RMD families (allelic overlap)
19380584	Mechanism	Golgi retention, dominant-negative, MG53–dysferlin–Cav3 repair defect
17039257	Mechanism	Myostatin/Smad2/p21 atrophy; rescued by myostatin inhibition
11786420	Mechanism	Caveolae loss + T-tubule disorganization in human muscle
14633633	Mechanism	R26Q impairs PFK-M membrane recruitment (metabolic branch)
15318349	Supporting	Incomplete penetrance (asymptomatic at 86); variable expressivity
19584897	Supporting	Same mutation → different phenotypes (modifiers/environment)
39370631	Diagnostic	Reduced caveolin-3 IHC; differential with immune-mediated RMD
30028203	Model	Cav3 KO disrupts T-tubules
10931944	Model	Overexpression → Duchenne-like dystrophy; dosage sensitivity
17275750	Pleiotropy	Gain-of-function late INa → LQT9/SIDS (other alleles)
23541953	Pleiotropy	Cav3–nNOS–SCN5A S-nitrosylation mechanism
24021552	Pleiotropy	Low cardiac penetrance of skeletal alleles (T78M no LQTS)

Limitations and Knowledge Gaps

1. **Thin phenotype-specific evidence base.** The Tateyama distal-myopathy subtype rests on a handful of case reports; no natural-history cohort, prevalence estimate, or quality-of-life study exists specifically for it.
2. **Genotype–phenotype gap.** Why the identical c.80G>A allele produces distal myopathy vs RMD vs hyperCKemia vs asymptomatic status is unexplained; **modifier genes remain unidentified**.
3. **Distribution not modeled.** No animal/cell model reproduces the distinctive hand/foot-restricted atrophy; the mechanistic link between generalized caveolin-3 loss and this focal distribution is unknown.
4. **Metabolic branch inferred.** The PFK-M/glycolysis contribution (PMID: 14633633) is in vitro and its clinical weight is uncertain.
5. **No human therapeutic data.** Myostatin inhibition is validated only in mice.
6. **Diagnostic overlap** with immune-mediated and *CAVIN1/PTRF*-related RMD can cause misclassification if serology/second-gene testing is omitted.

Proposed Follow-up Experiments / Actions

1. **Genotype–phenotype modifier study:** WGS/transcriptomic comparison of c.80G>A carriers across phenotypes (distal myopathy vs RMD vs asymptomatic) to map modifier loci.
2. **Patient-derived iPSC myotubes** carrying p.Arg27Gln to quantify caveolae loss, membrane-repair kinetics (laser-wounding assay), and myostatin/Smad2 activation, and to test myostatin-pathway inhibitors in vitro.
3. **Natural-history registry** for CAV3 distal myopathy capturing MRI progression, CK trajectory, and PROMIS/SF-36 quality-of-life metrics.
4. **Myostatin-inhibition translational rationale** building on the mouse rescue (PMID: 17039257).
5. **Systematic serology** (anti-cavin-4/MURC) and *CAVIN1/PTRF* sequencing in all reduced-caveolin-3 cases to exclude acquired and secondary mimics.
6. **Cardiac screening protocol** stratified by specific *CAV3* allele (reassurance for Arg27Gln; ECG/Holter for gain-of-function alleles).

Report compiled from a 5-iteration autonomous investigation: 10 confirmed findings, 34 papers reviewed. Evidence types span human clinical case reports/cohorts, mouse models (knockout, dominant-negative transgenic, overexpression), and in vitro cell studies.